Welded steel workbench

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Make it yours

Length 1500 mm · Depth 700 mm · Work height 900 mm · Rated load 300 kg · Gap between top bars 10 mm · Lower shelf yes

1 Checks

	Check	Result
✓	**Every joint can be made as drawn** Each end is a mitre, a butt onto a face, a flush cut or a free end; nothing overhangs.	`24 joints`
✓	**Nothing floating** Every member, plate and part touches what holds it.	`all connected`
✓	**No clashes** No member passes through another away from a joint, and no plate through a member it is not notched around.	`none`
✓	**Every piece fits the stock** Cut from standard lengths with a trim and a kerf per cut.	`longest 1500 mm`
✓	**Every piece can be clamped and cut** No piece shorter than 32 mm on its short side.	`shortest 620 mm`
✓	**Rated load on the top: strength** Yield ÷ peak stress with 337 kg spread over the top frame, with the bars, plus self weight.	`safety factor 4.46 (need 2), worst A (leg)`
✓	**Rated load on the top: deflection** Largest movement of the loaded members.	`1.7 mm (limit 4.9 mm)`
✓	**Point load at a front corner: strength** Yield ÷ peak stress with 150 kg on one front corner, plus self weight.	`safety factor 16.78 (need 2), worst A (leg)`
✓	**Shoved sideways at the top: strength** Yield ÷ peak stress with 400 N pushing the top end-on, plus self weight.	`safety factor 9.95 (need 2), worst A (leg)`
✓	**Shoved front to back at the top: strength** Yield ÷ peak stress with 400 N pushing the top front edge, plus self weight.	`safety factor 11.81 (need 2), worst A (leg)`
✓	**Loaded lower shelf: strength** Yield ÷ peak stress with 100 kg spread over the shelf, plus self weight.	`safety factor 19.42 (need 2), worst C (long rail)`
✓	**Loaded lower shelf: deflection** Largest movement of the loaded members.	`0.3 mm (limit 5.8 mm)`
✓	**Does not tip over** Free-standing it would tip with 300 N sideways at 900 mm high, so it is fixed down.	`anchored (step 9, an M8 floor anchor through the base of every levelling foot); free-standing factor 1.13`
✓	**Every part has a step** Each piece is cut, fitted or bought in a numbered step.	`9 steps`
✓	**Top bars: strength** Each bar spans 487 mm between cross rails: 100 kg on the middle of one span, or its share of the rated load.	`safety factor 2.83 (need 2), sags 0.8 mm under 100 kg`
✓	**Shelf bars: strength** Each shelf bar spans 660 mm between the lower rails, with 100 kg on its middle.	`safety factor 2.09 (need 2)`

✓	**Gaps between the top bars** At least the gap asked for, evened out so the outer bars sit flush with the front and back.	14 bars, 10.8 mm gaps (asked 10 mm)
✓	**Work height** Standing work: about wrist height for a vise, elbow height for fine work.	900 mm
✓	**Room above the shelf** At least 300 mm between the shelf and the top frame for boxes and cases.	600 mm clear

2 Materials

1500 × 700 × 900 mm · 95 kg · 49 pieces · 4 plates · 20 h of work · ~$239 (priced lines) · Work height: 900 mm · Top: 14 steel bars

Stock

Section	Pieces	Cut length	Buy	Mass	Cost
40 × 40 × 2 square tube	14	12.8 m	3 × 6.00 m	30 kg	$72
40 × 20 × 2 rectangular tube, laid flat	35	35.7 m	7 × 6.00 m	63 kg	$143

Plate and sheet

Mark	Part	Size	Qty	Step	Cost
P1	Foot plate	40 × 40 × 6 steel	4	7	$2.40

Parts to buy

Part	Spec	Qty	Step	Cost
Hex nut, M12	plain steel, welded to the foot plate	4	7	$0.60
Levelling foot, M12, bolt-down base	40 mm base with an anchor hole, M12 stud at least 60 mm, 500 kg rating	4	9	$16
Floor anchor, M8	sleeve or wedge anchor, to suit the levelling foot base hole	4	9	$4.80

Consumables

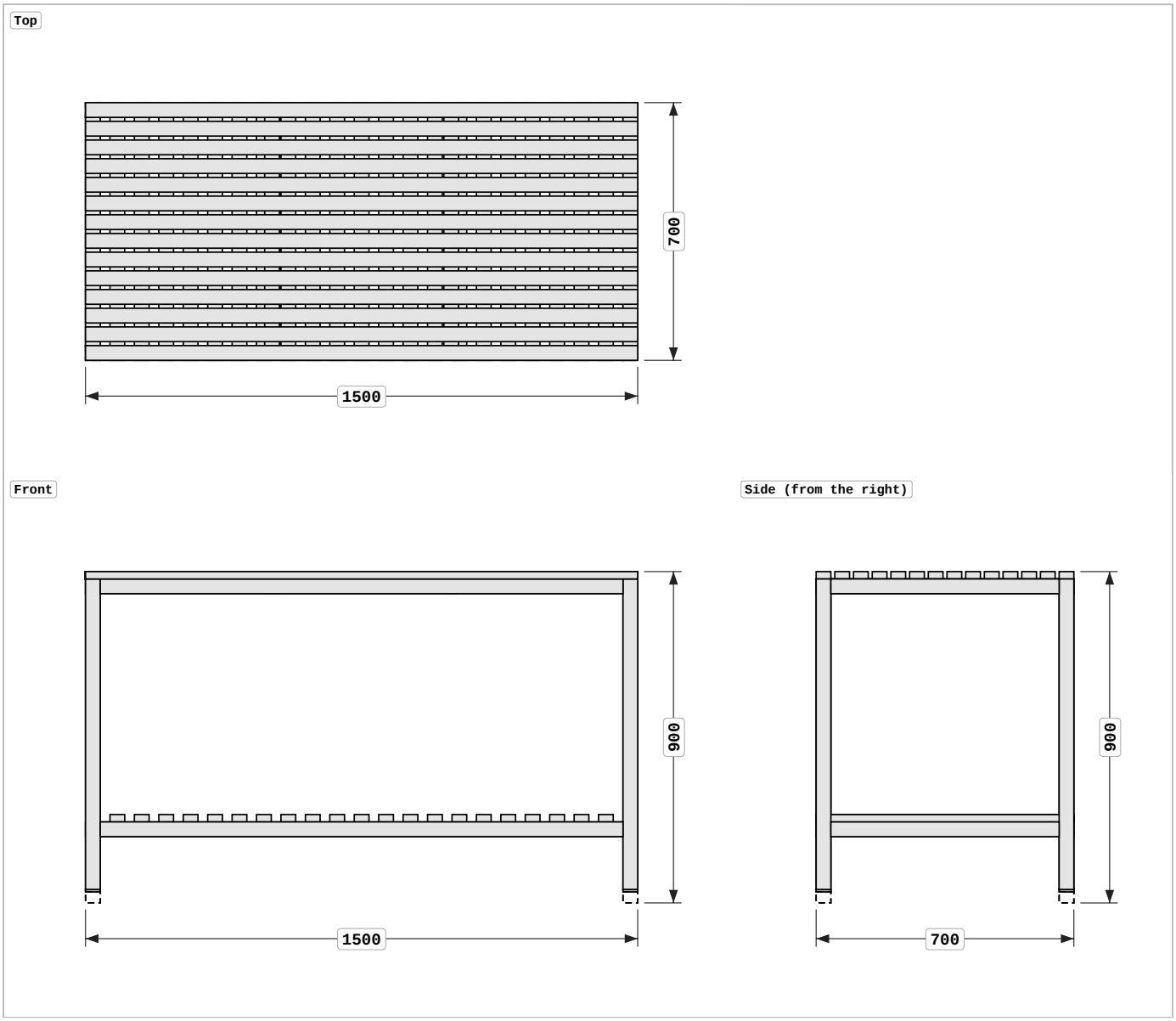
- MIG wire, 0.8 mm · 1 kg spool
- Shielding gas, Ar/CO2 · one small cylinder

- Cut-off discs, 1 mm · 5
- Flap discs, 80 grit · 3
- Primer and enamel · 1 L each
- Degreaser and rags

Tools

- MIG welder
- Hammer drill and masonry bit for the anchors
- Chop saw or angle grinder with cut-off discs
- Angle grinder with flap disc
- Welding table or flat floor
- Square, tape, scriber, spirit level
- Six or more clamps
- Drill, 13 mm bit, M12 tap
- A spacer block for the bar gaps

3 Drawings



4 Legend

Units	millimetres, kilograms, degrees
Datum	x from the left end of the top, y from the front edge, z up from the floor
Piece marks	A, B, C… one letter per different piece; x4 after it means cut four the same. Mirror images share a mark.
Cut marks	From one square end, mark each edge at the two numbers, join the marks around the section, cut on the lines. Angles are for reference.
Edges	named by where they face on the finished build (top-front, bottom-right…)
Joints	mitre = same section, on the bisector · butt onto X = square onto the face of X · flush with X = cut level with X's outer face · then weld
Long point	the overall length of the piece, tip to tip
Dimensions	jig drawings and joint tables give centre lines (℄), where the members' centres cross; the general arrangement gives outside sizes
Tolerance	cut ±1 mm · tacked ±1 mm · finished ±2 mm · diagonals equal within 3

5 Order

Cut everything, tack the end frames, join them, weld out, weld the bars on, then feet, paint and load test.

Fabricate

1. Cut, deburr, foot plates 3 h
2. Two end frames (tack) 2 h
3. Join the end frames (tack) 2 h
4. Cross rails (tack) 1 h
5. Weld out the frame 3 h
6. Top and shelf bars 4 h
7. Foot plates 1 h

Assemble

9. Feet, anchors and load test 2 h

Finish

8. Clean and paint 2 h

6 Steps

1 Cut, deburr, foot plates

3 h

- [ ] 1. Cut every piece from the cutting plan, ends square.
- [ ] 2. Deburr each end. Write its piece mark on it.
- [ ] 3. Count against the cut list: 4 legs (A), 45 rails.
- [ ] 4. 4 foot plates 40 × 40 from 6 mm plate. Drill 13 mm in the centre.

✓ Every piece ±1 mm of its long point.

! Glasses, gloves, ear protection. Bare or oiled tube only, never galvanised.

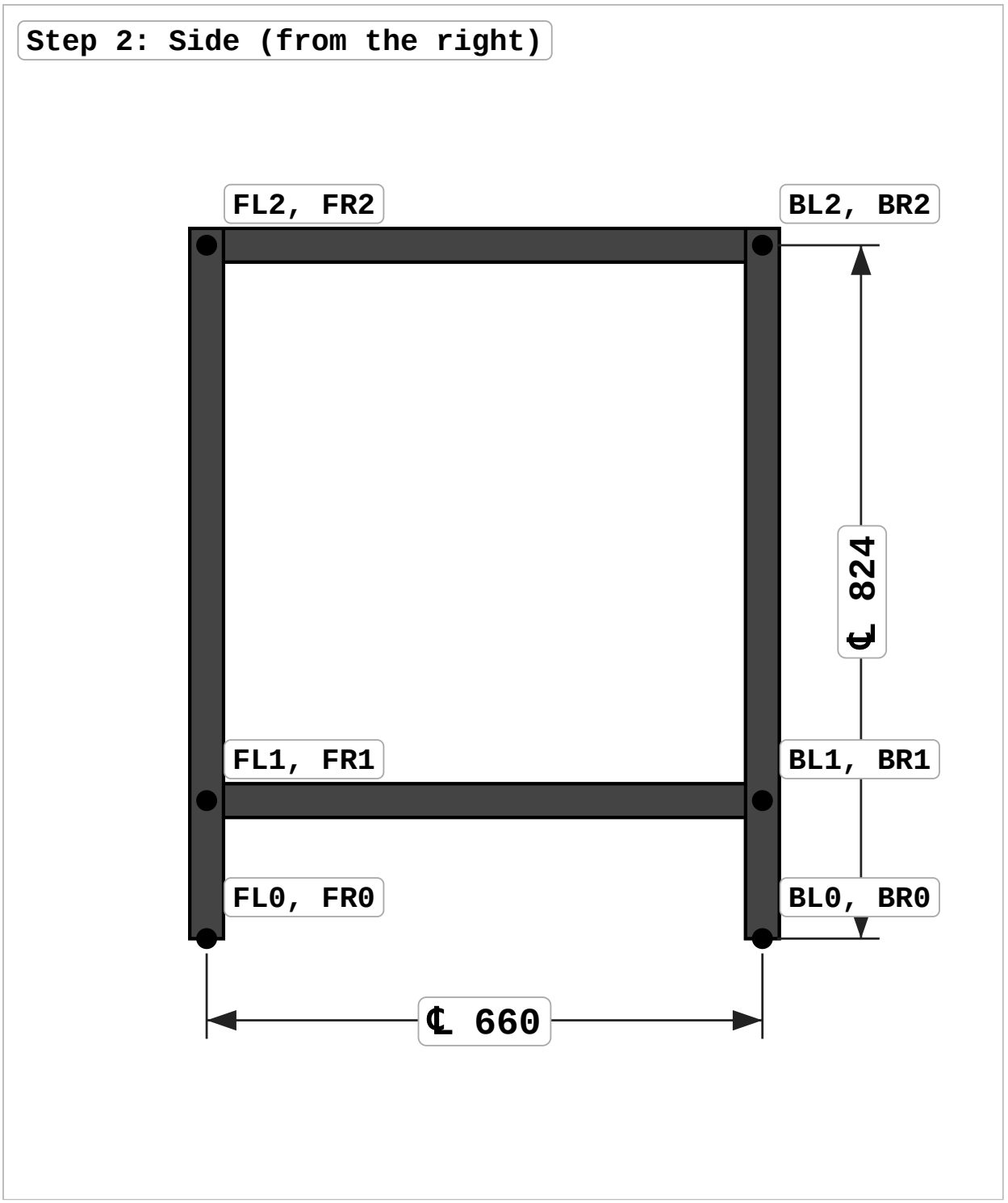
2 Two end frames (tack)

2 h

- [ ] 1. Two legs (A) flat on the table, parallel.
- [ ] 2. Short rails (B) between them: one flush with the leg tops, one 164 up to its centre line.
- [ ] 3. Outside width 700. Clamp square. Tack each joint on two faces.
- [ ] 4. Build the second frame on top of the first.

✓ Diagonals equal ±2 mm. Both frames the same ±1 mm.

Jig


Step 2: Side (from the right)
FL2, FR2
BL2, BR2
℄ 824
FL1, FR1
BL1, BR1
FL0, FR0
BL0, BR0
℄ 660

Joint positions

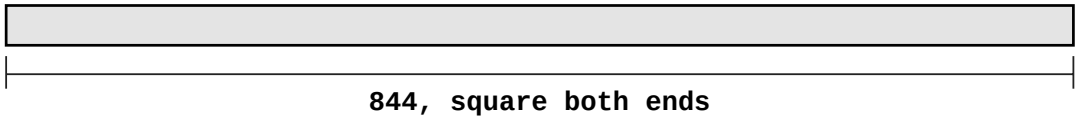
Point	x	y	z
FL0	20	20	36
FR0	1480	20	36
BL0	20	680	36
BR0	1480	680	36
FL1	20	20	200
FR1	1480	20	200
BL1	20	680	200
BR1	1480	680	200
FL2	20	20	860
FR2	1480	20	860
BL2	20	680	860
BR2	1480	680	860

Pieces: A x4, B x4 (of 6)

A x4 — **844 mm**

Leg · SHS 40×40×2 · step 2 — long point

`leg-FL, leg-FR, leg-BL, leg-BR`

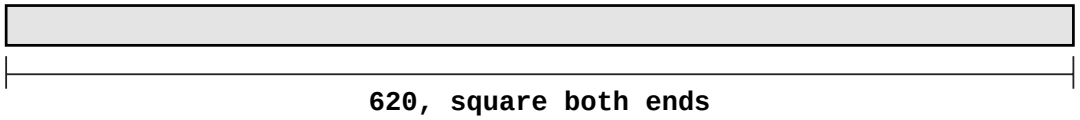


Left end: onto P1 (foot plate) · Right end: flush with B (short rail)

B x6 — **620 mm**

Short rail · SHS 40×40×2 · step 2 — long point

`end-L-top, end-L-low, end-R-top, end-R-low, cross-top-1, cross-top-2`



Left end: butt onto A (leg) or C (long rail) · Right end: butt onto A (leg) or C (long rail)

3 Join the end frames (tack)

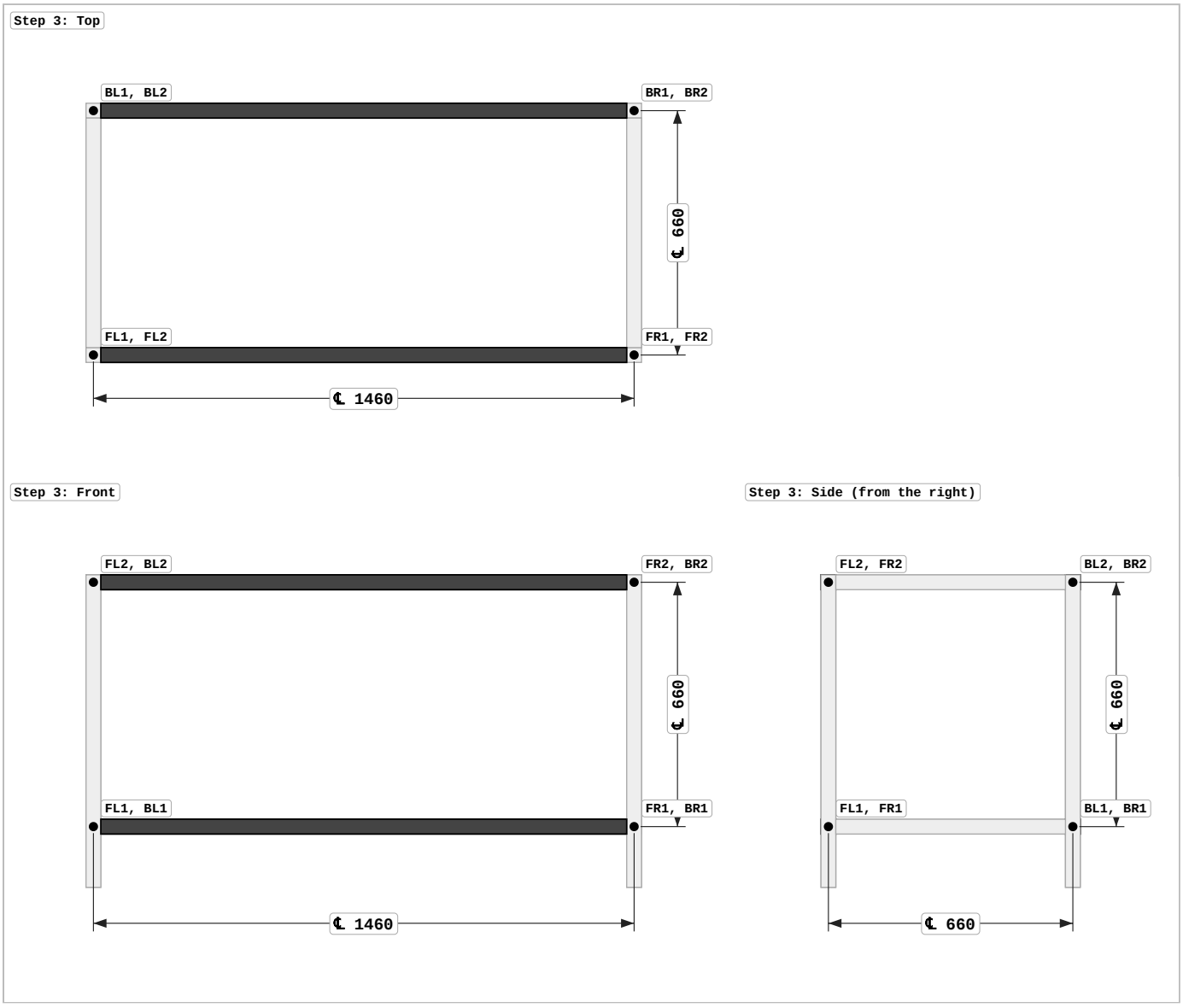
2 h

- [ ] 1. Stand both end frames up. Clamp each to a square.
- [ ] 2. Long rails (C) between the frames: top ones flush with the leg tops, lower ones level with the lower end rails.
- [ ] 3. Outside length 1500, top and bottom.
- [ ] 4. Tack the top rails, then the lower ones.
- [ ] 5. Equal the top diagonals, then tack every joint on two faces.

✓ Top diagonals equal ±3 mm. Legs plumb ±1 mm. Stands on all 4 legs without rocking.

! No welding until step 5.

Jig



Joint positions

Point	x	y	z
FL1	20	20	200
FR1	1480	20	200
BL1	20	680	200
BR1	1480	680	200
FL2	20	20	860
FR2	1480	20	860
BL2	20	680	860
BR2	1480	680	860

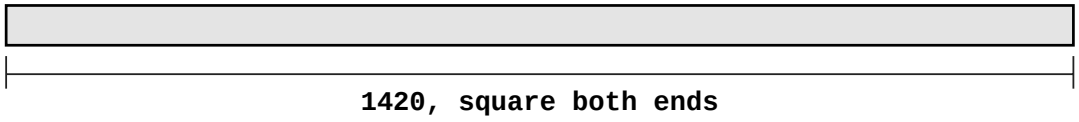
Pieces: C x4

C x4 — **1420 mm**

Long rail · SHS 40×40×2 · step 3

long point

`long-F-top, long-B-top, long-F-low, long-B-low`



Left end: butt onto A (leg) · Right end: butt onto A (leg)

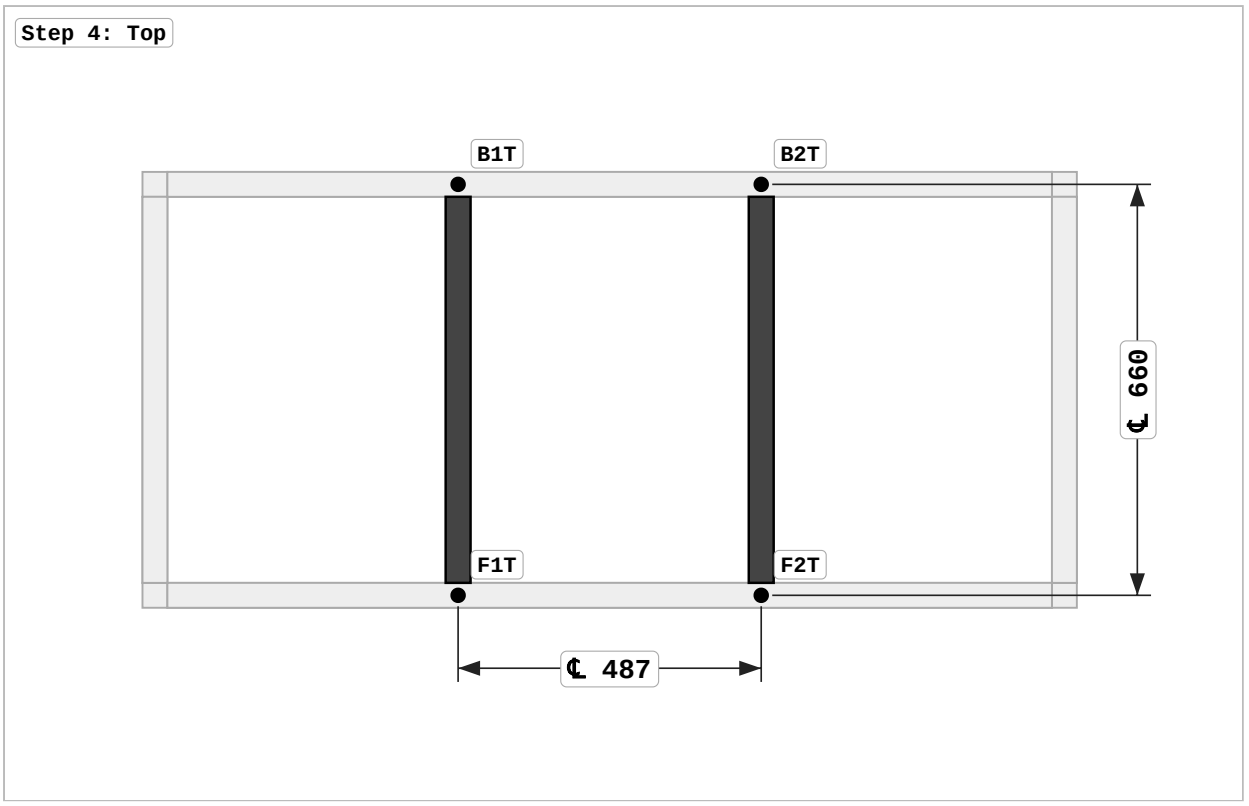
4 Cross rails (tack)

1 h

- [ ] 1. Mark the top long rails at 507, 993 from the left end.
- [ ] 2. A short rail (B) at each mark, flush with the top. Tack both ends.

✓ Flush with the top ±1 mm. 487 apart centre to centre ±2 mm.

Jig



Joint positions

Point	x	y	z
F1T	507	20	860
F2T	993	20	860
B1T	507	680	860
B2T	993	680	860

Pieces: B x2 (of 6)

B: cut sheet in step 2

5 Weld out the frame

3 h

- [ ] 1. Check the top diagonals and the plumb first.
- [ ] 2. Weld out in short runs, diagonally opposite corners in turn.
- [ ] 3. Top frame, then lower frame, then cross rails. Clamped until cold.
- [ ] 4. Grind the welds on the rail tops flush.

✓ Top diagonals still equal ±3 mm. A straight edge across the rails shows no gap over 1 mm.

! Ventilate: fumes from mill scale and paint are toxic.

6 Top and shelf bars

4 h

- [ ] 1. 14 top bars (D) flat on the frame, lengthwise, ends flush.
- [ ] 2. Outer two flush front and back. 10.8 mm gaps between: cut a spacer block that wide.
- [ ] 3. Tack each bar at both ends, then to every cross rail.
- [ ] 4. 21 shelf bars (E) across the lower long rails, ends flush, 26 mm gaps. Tack both ends.
- [ ] 5. Stitch-weld every crossing, 25 mm each side, middle bar outward (98 crossings).
- [ ] 6. Grind proud welds flush.

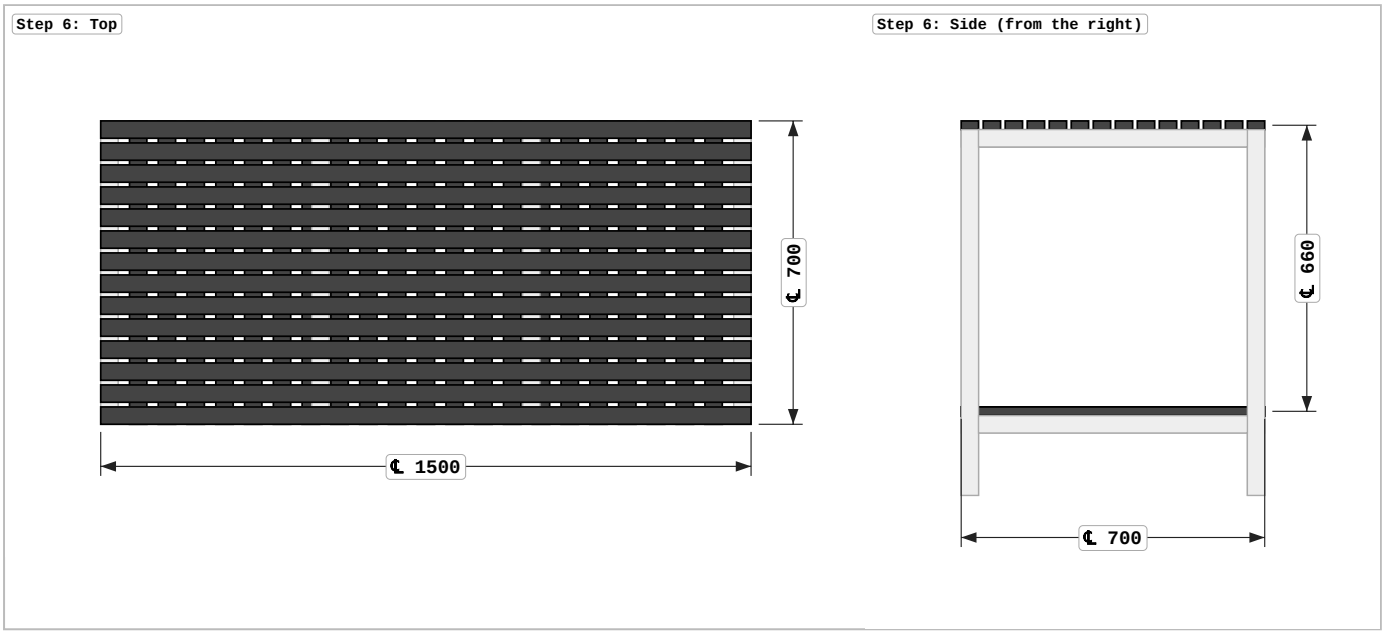
✓ Gaps equal ±1 mm. A straight edge across the bars rocks nowhere.

! Never weld one bar end to end: the top bows.

Bars

Where	Bars	Runs	Gap	First centre line
Top	D x14	left to right	10.8	20 from the front
Shelf	E x21	front to back	26	86 from the left end

Jig

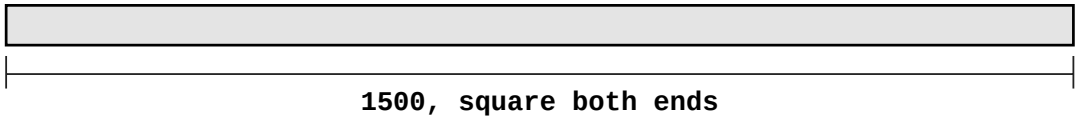


Pieces: D x14, E x21

D x14 **1500 mm**

Bar · RHS 40×20×2 · step 6 long point

top-bar-1, top-bar-2, top-bar-3, top-bar-4, top-bar-5, top-bar-6, top-bar-7, top-bar-8, …and 6 more

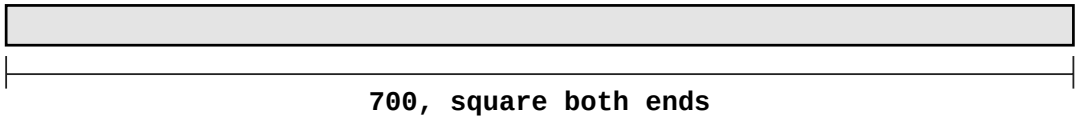


Left end: free end · Right end: free end

E x21 **700 mm**

Shelf bar · RHS 40×20×2 · step 6 long point

shelf-bar-1, shelf-bar-2, shelf-bar-3, shelf-bar-4, shelf-bar-5, shelf-bar-6, …and 15 more



Left end: free end · Right end: free end

7 Foot plates

1 h

- [ ] 1. Weld an M12 nut over each plate's hole. A bolt through both keeps it straight.
- [ ] 2. Bench on its back (two people). A plate centred under each leg, nut inside. Weld all round.
- [ ] 3. Run a bolt in and out of each nut to clear spatter.

✓ A bolt screws into every nut by hand, straight.

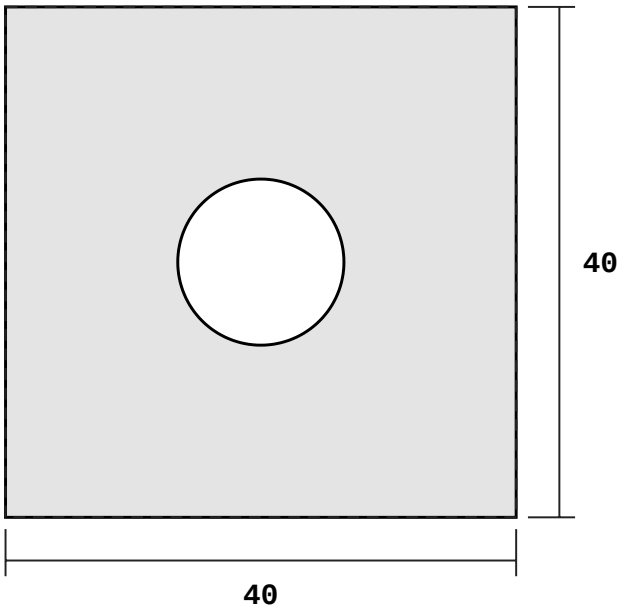
Parts

- Hex nut, M12 ×4 plain steel, welded to the foot plate

Plates

P1 x4

Foot plate · 6 mm steel · step 7



1 hole ⌀13 mm, centres from the lower-left corner: (20, 20).

Dashed: where a member meets its face (welded on, or under it), from the lower-left corner: (0, 0).

Weld an M12 nut over the hole on the top face first, then weld the plate centred under the leg, all round.

8 Clean and paint

2 h

- [ ] 1. Degrease. Wire-brush the welds.
- [ ] 2. Prime. Two coats of enamel, into the bar gaps. Dry overnight.
- [ ] 3. Or leave the bar tops bare and oiled.

✓ No bare steel but the oiled tops. Dry to the touch.

9 Feet, anchors and load test

2 h

- [ ] 1. Levelling foot in each leg, 30 under the plate.
- [ ] 2. Level the top both ways with the feet. Lock the nuts.
- [ ] 3. Drill and set the M8 anchors at the foot holes. Bolt every foot down (4).
- [ ] 4. Load-test: a quarter at a time up to 300 kg over the top. Leave it an hour.

✓ Level ±1 mm. Does not move when pushed at the top edge. No sag, no cracked weld.

Parts

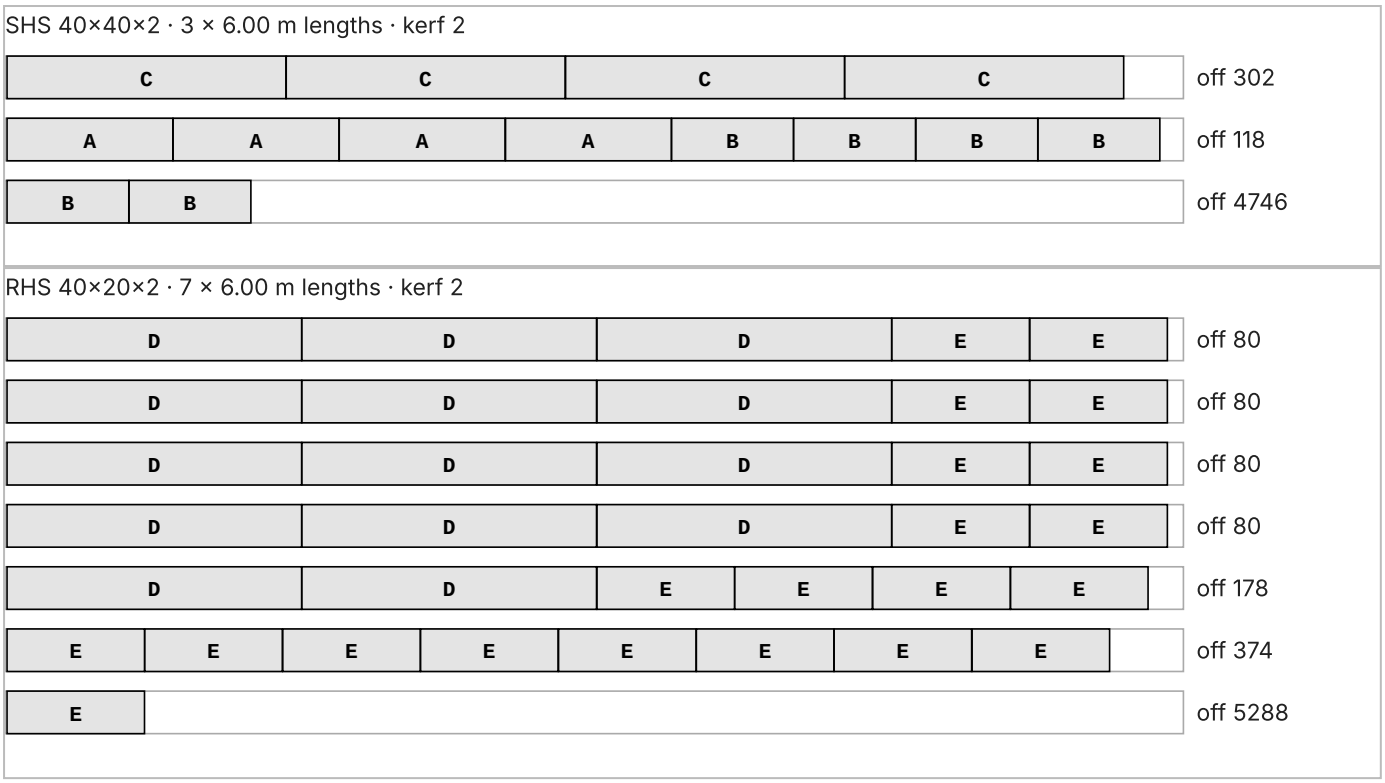
- Levelling foot, M12, bolt-down base x4 40 mm base with an anchor hole, M12 stud at least 60 mm, 500 kg rating
- Floor anchor, M8 ×4 sleeve or wedge anchor, to suit the levelling foot base hole

7 Cut list

Mark	Qty	Section	Long point	Left end	Right end	Step
A Leg	4	SHS 40×40×2	844	square · onto P1 (foot plate)	square · flush with B (short rail)	2
B Short rail	6	SHS 40×40×2	620	square · butt onto A (leg) or C (long rail)	square · butt onto A (leg) or C (long rail)	2
C Long rail	4	SHS 40×40×2	1420	square · butt onto A (leg)	square · butt onto A (leg)	3
D Bar	14	RHS 40×20×2	1500	square · free end	square · free end	6
E Shelf bar	21	RHS 40×20×2	700	square · free end	square · free end	6

Cutting plan

Each bar is one stock length; cut the pieces left to right.



Welded steel workbench · v1. Generated by the blueprint skill. Check every dimension against your own stock before you cut, wear eye and hand protection, and load-test the build gently before trusting it.